Gilles Eerlings

PhD Researcher in Machine Learning & Human-Centered Artificial Intelligence

gilleseerlings99@gmail.com | linkedin.com/in/gilles-eerlings | Google Scholar | gilleseerlings.be

Summary

PhD researcher working at the intersection of machine learning, model multiplicity, and human-AI collaboration. I develop methods for efficiently exploring sets of similarly performing models and investigate how their disagreement can be communicated to support human decision-making.

Experience

PhD Researcher in Machine Learning & Human-Centered AI
UHasselt — Digital Future Lab
Sep 2024 — Present
Diepenbeek, Belgium

- Research model multiplicity in machine learning, spanning efficient Rashomon-set exploration in deep neural networks and the use of model disagreement in human-AI decision-making.
- Developed **DIVERSE**, a gradient-free framework combining FiLM-based neural-network modulation with CMA-ES to explore Rashomon sets without retraining model weights; **first-author publication at ICLR 2026**.
- Evaluated DIVERSE across MNIST, CIFAR-10, and PneumoniaMNIST using MLP, VGG-16, and ResNet-50 architectures, benchmarking against retraining- and dropout-based approaches; released the implementation and experiment pipeline as open-source software.
- Develop interactive methods for communicating model disagreement and decision stability in AI-assisted decision-making.
- Design and conduct controlled human-subject studies to investigate how model multiplicity affects reliance on AI, confidence, and decision-making.

Research Intern
UHasselt — Digital Future Lab
Aug. 2023 — Sep. 2023
Diepenbeek, Belgium (Hybrid)

- Explored gesture-based interaction approaches for a multimodal digital assistant supporting assembly operators.
- Built and compared proof-of-concept **hand-gesture recognition approaches** using smartwatch sensor data and Leap Motion hand tracking.
- Developed a Wear OS application in Kotlin to collect and label smartwatch sensor data, creating a custom gesture dataset for machine-learning experiments.
- Implemented and evaluated recurrent neural network models for gesture recognition using Python and TensorFlow, with data processing and analysis in Pandas and NumPy.

Web Developer (Part-time)
KU Leuven – Department of Cellular and Molecular Medicine
Aug. 2022 — Mar. 2023
Leuven, Belgium (Remote)

- Redesigned the departmental website and research group pages, restructuring navigation and content to improve usability, accessibility, and maintainability.
- Implemented the redesigned website and migrated existing content using the Plone content management system.
- Collaborated with researchers and administrative staff to gather requirements, iterate on the redesign, and validate changes.

Research Intern
PXL Smart ICT
Mar. 2021 — Aug. 2021
Hasselt, Belgium

- Re-engineered **dAIsy**, a rule-based academic-paper recommender, into an NLP-based chatbot using Rasa and its DIET model for intent classification and named-entity recognition.
- Integrated **Transformer-based document summarization** using AllenAI's Longformer Encoder-Decoder (LED) through Hugging Face, enabling automatic summarization of academic papers.
- Developed an asynchronous summarization pipeline using FastAPI, RabbitMQ, Celery, and Redis to support concurrent requests, deploying the system across seven communicating Docker containers.
- Implemented a content-based paper recommender using word embeddings and nearest-neighbor similarity to recommend relevant research.

- Built a React frontend exposing model predictions and confidence scores and visualizing summarization attention through heatmaps.

Education

PhD in Computer Science — UHasselt — Hasselt University
Sep. 2024 — Expected Sep. 2028
Digital Future Lab · Supervised by Kris Luyten and Gustavo Rovelo Ruiz

MSc in Computer Science — UHasselt — Hasselt University
Sep. 2021 — Sep. 2024
Thesis: Many Minds, One Decision: Leveraging Multiple AI Models for Informed Decision-Making · Supervised by Kris Luyten, Jori Liesenborgs, and Gustavo Rovelo Ruiz

BSc in Applied Computer Science — PXL University of Applied Sciences and Arts
Sep. 2018 — Jun. 2021
Specialization: Artificial Intelligence
Thesis: XAI: dAIsy -- Natural Language Powered Chatbot

Technical Skills

Programming	Python, JavaScript, SQL, Bash, Java
ML & Data	TensorFlow, Keras, scikit-learn, PyTorch, NumPy, Pandas
Software Development	FastAPI, ReactJS, Tailwind CSS, Docker, Git
Computing & Research Tools	Linux, SLURM, Jupyter, MLflow, LaTeX

Teaching & Supervision

Teaching Assistant — UHasselt — Hasselt University
2024 — Present
- Teaching Assistant for Bachelor's and Master's courses in Introduction to AI and Machine Learning, Machine Learning, Human-AI Interaction, Computational Optimization Techniques, and Advanced Object-Oriented Programming.
- Designed and supervised course assignments and projects
- Led exercise and lab sessions, providing hands-on support during practical coursework
- Conducted individual feedback sessions and participated in grading and evaluation of assignments
- Assisted in the preparation of examination materials and the assessment of oral examinations

Thesis Supervision — Bachelor's and Master's Theses — UHasselt — Hasselt University
2024 — Present
- Daily supervisor for **more than 10 Bachelor's and Master's theses**
- Supervised thesis topics in machine learning, explainable AI, trustworthy AI, and human-centered AI
- Provided methodological, technical, and conceptual guidance throughout the thesis process
- Assisted in the evaluation of written theses and oral defenses

Publications

Title anonymized *(under review)*
Gilles Eerlings, Seppe Vandenberk, Sebe Vanbrabant, Gustavo Rovelo Ruiz, Kris Luyten
The ACM CHI Conference on Human Factors in Computing Systems, CHI — 2027

From Embeddings to Exploration: Engineering Interactive Latent Space Visualizations for AI Model Sensemaking
Sebe Vanbrabant, Jarne Tyhs, Gilles Eerlings, Kris Luyten, Gustavo Rovelo Ruiz, Davy Vanacken
The 18th ACM SIGCHI Symposium on Engineering Interactive Computing Systems, EICS — 2026

Reclaiming Knowledge Structuring from Automation: Human-AI Co-Engineering of KC Models
Seppe Vandenberk, Gilles Eerlings, Gustavo Rovelo Ruiz, Davy Vanacken

The 4th Workshop on Engineering Interactive Systems Embedding AI Technologies — 2026

DIVERSE: Disagreement-Inducing Vector Evolution for Rashomon Set Exploration
Gilles Eerlings, Brent Zoomers, Jori Liesenborgs, Gustavo Rovelo Ruiz, Kris Luyten
The Fourteenth International Conference on Learning Representations, ICLR — 2026

ECHO: Enhancing Conversational Explainable AI through Tool-Augmented Language Models
Sebe Vanbrabant, Gilles Eerlings, Gustavo Rovelo Ruiz, Davy Vanacken
The 17th ACM SIGCHI Symposium on Engineering Interactive Computing Systems, EICS — 2025

AI-Spectra: A Visual Dashboard for Model Multiplicity to Enhance Informed and Transparent Decision-Making
Gilles Eerlings, Sebe Vanbrabant, Jori Liesenborgs, Gustavo Rovelo Ruiz, Davy Vanacken, Kris Luyten
Engineering Interactive Computer Systems. EICS 2024 International Workshops, Springer — 2024

Opportunities and Challenges of Model Multiplicity in Interactive Software Systems
Kris Luyten, Gilles Eerlings, Jori Liesenborgs, Gustavo Rovelo Ruiz, Sebe Vanbrabant, Davy Vanacken
2nd Workshop on Engineering Interactive Systems Embedding AI Technologies — 2024

Presentations & Posters

DIVERSE: Disagreement-Inducing Vector Evolution for Rashomon Set Exploration
International Conference on Learning Representations (ICLR) — Rio de Janeiro, Brazil — 2026

AI Needs More Opinions - Embracing the Rashomon Perspective for Fairer and Transparent Decision-Making
Flanders AI Research Day — Ghent, Belgium — 2025

AI-Spectra: A Visual Dashboard for Model Multiplicity to Enhance Informed and Transparent Decision-Making
Flanders AI Research Day — Ghent, Belgium — 2024

Academic Service

Peer Reviewing
UIST 2026, AIES 2026, NeurIPS 2026, ICLR 2027